

**Thm.** Let  $f$  be analytic in the disk  $|z - z_0| < R$ ,  
 and  $\{z_n\}$  be a sequence of distinct points converging to  $z_0$ .  
 If  $f(z_n) = 0$  for all  $n \in \mathbb{N}$ , then  $f \equiv 0$  in  $|z - z_0| < R$ .

**Proof.** Since  $f$  is analytic, the Taylor expansion

$$f(z) = a_0 + \sum_{k=1}^{\infty} a_k (z - z_0)^k$$

is valid in  $|z - z_0| < R$ . Since  $f$  is continuous,

$$\lim_{n \rightarrow \infty} f(z_n) = f(z_0) = a_0.$$

so  $a_0 = 0$ . So, write

$$f(z) = (z - z_0) \cdot \left( a_1 + \sum_{k=2}^{\infty} a_k (z - z_0)^{k-1} \right)$$

let  $z = z_n$ , so

$$\lim_{n \rightarrow \infty} \frac{f(z_n)}{z_n - z_0} = a_1 = 0, \quad \text{inductively, } a_0 = a_1 = \dots = 0.$$

**Corollary.** If  $f$  is analytic at  $z_0$ , then there is  $r > 0$  such that

$$f(z) \neq 0 \quad \text{for all } 0 < |z - z_0| \leq r$$

or  $f \equiv 0$

**Proof.** If no such  $r > 0$  exists, then for each  $n \in \mathbb{N}$ , there is  $z_n \in B_{1/n}^c(z_0)$  s.t.  
 $f(z_n) = 0$ .

Clearly  $z_n \rightarrow z_0$ , so  $f \equiv 0$ .

### The Uniqueness Theorem.

Let  $f$  be analytic in a domain  $D$ , and that  $\{z_n\}$  is a sequence of distinct points, converging  $z_0 \in D$ .

If  $f(z_n) = 0$  for all  $n \in \mathbb{N}$ , then  $f \equiv 0$  throughout  $D$ .

Proof. Define two sets:

$$A := \{a \in D \mid f \equiv 0 \text{ on } B_r(a) \text{ for some } r > 0\}$$

$$B := \{a \in D \mid f \neq 0 \text{ on } B_r(a) \setminus \{a\} \text{ for some } r > 0\}$$

[Lemma. Let  $f$  be an analytic in the  $B_R(z_0)$ .

If the set  $Z(f) = f^{-1}[0]$  has a limit point in  $B_R(z_0)$ , then  $f \equiv 0$  on  $B_R(z_0)$ .

Proof. Let  $z_0 \in B_R(z_0)$  be the limit point of  $Z(f)$ , by the hypothesis. Since  $f$  is analytic, the representation

$$f(z) = a_0 + \sum_{k=1}^{\infty} a_k (z - z_0)^k$$

is valid, and since  $z_0$  is the limit point of  $Z(f)$ , there is a sequence  $\{z_n\}$  in  $Z(f)$ , converging to  $z_0$ . Therefore,

$$\lim_{n \rightarrow \infty} f(z_n) = f(z_0) = a_0$$

is zero. Then, now,

$$f(z) = \sum_{k=1}^{\infty} a_k (z - z_0)^k = (z - z_0) \cdot \left( a_1 + \sum_{k=2}^{\infty} a_k (z - z_0)^{k-1} \right)$$

And, similarly, we can construct  $\{z_n\}$  not containing  $z_0$ , so that

$$\frac{f(z_n)}{z_n - z_0} \rightarrow 0, \text{ so } a_1 \text{ is also } 0. \text{ Inductively, we obtain } a_n = 0 \text{ for all } n \in \mathbb{N}. \quad \square$$

[Corollary. If  $f$  is analytic at  $z_0$ , then either

$f \equiv 0$  in some  $B_r(z_0)$  or  $f \neq 0$  in some  $B_{r'}(z_0) \setminus \{z_0\}$ .

Proof. If there is no such  $r'$  exists, that is, for all  $r' > 0$ , there is  $z \in B_{r'}(z_0)$  s.t.  $f(z) = 0$ , then  $f \equiv 0$  for some neighbor, by the Lemma above.

[Uniqueness Theorem.

Let  $f$  be an analytic in a domain  $D$ , and  $Z(f)$  has a limit point in  $D$ .

Then  $f \equiv 0$  on  $D$ .